

Exercițiul 1.

(3 puncte)

Pentru $z \in \mathbb{C}$, notăm $z = x + iy$, $x, y \in \mathbb{R}$ și avem identificarea $z \simeq (x, y)$. Considerăm funcția $u : \mathbb{C} \rightarrow \mathbb{R}$, $u(x, y) = x^4 + y^4 - 6x^2y^2 - x^3y + xy^3$. Găsiți o funcție olomorfă $f : \mathbb{C} \rightarrow \mathbb{C}$ pentru care $\operatorname{Re}(f) = u$.

Exercițiul 2.

(3 puncte)

Folosind teorema reziduurilor, calculați următoarea integrală:

$$\int_{-\infty}^{\infty} \frac{x^2 - 1}{x^4 + 2x^2 + 1} dx.$$

Exercițiul 3.

(3 puncte)

Notăm $\Omega =$ semiplanul superior H^+ din care scoatem segmentul vertical $i \cdot [0, 1]$. Găsiți un șir de aplicații conforme bijective care transformă Ω în discul $D(0, 1)$.

Exercițiul 4.

(3 puncte)

Considerăm $b \in \mathbb{C}$ cu $|b| < 1$. Demonstrați că $f(z) = z^3 + 3z^2 + bz + b^2$ are exact o rădăcină în afara discului $|z| < 1$.

Indicație: Folosim teorema lui Rouché.

Exercițiul 1:

$$u: \mathbb{C} \rightarrow \mathbb{R}, \quad u(x, y) = x^4 + y^4 - 6x^2y^2 - x^3y + xy^3$$

Verificăm întâi dacă funcția u este armonică.

$$\begin{aligned} \frac{\partial u}{\partial x} &= 4x^3 - 6y^2 \cdot 2x - y \cdot 3x^2 + y^3 = \\ &= 4x^3 - 12xy^2 - 3x^2y + y^3 \end{aligned}$$

$$\frac{\partial^2 u}{\partial x^2} = 12x^2 - 12y^2 - 3y \cdot 2x = 12x^2 - 12y^2 - 6xy$$

$$\begin{aligned} \frac{\partial u}{\partial y} &= 4y^3 - 6x^2 \cdot 2y - x^3 + x \cdot 3y^2 = \\ &= 4y^3 - 12x^2y - x^3 + 3xy^2 \end{aligned}$$

$$\frac{\partial^2 u}{\partial y^2} = 12y^2 - 12x^2 + 3x \cdot 2y = 12y^2 - 12x^2 + 6xy$$

$$\Delta u = \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 12x^2 - 12y^2 - 6xy + 12y^2 - 12x^2 + 6xy = 0 \Rightarrow$$

$\Rightarrow u$ este armonică pe $\mathbb{C}$
 $\mathbb{C}$ domeniu simplu conex $\Rightarrow$

$\Rightarrow$ există $f: \mathbb{C} \rightarrow \mathbb{C}$ olomorfa astfel încât $\operatorname{Re}(f) = u$

$$\begin{aligned} f'(z) = \frac{\partial f}{\partial z} &= \frac{\partial f}{\partial x} + i \frac{\partial f}{\partial y} = \frac{\partial u}{\partial x} + i \left(\frac{\partial u}{\partial y} - \left(\frac{\partial u}{\partial x} + i \frac{\partial u}{\partial y} \right) i \right) = \\ &= \left(4x^3 - 12xy^2 - 3x^2y + y^3 \right) + i \left(4y^3 - 12x^2y - x^3 + 3xy^2 \right) \quad \forall z = x + iy \in \mathbb{C} \end{aligned}$$

considerăm $y=0$, adică $z \in \mathbb{R}$:

$$f'(z) = 4z^3 - i(-z^3) = 4z^3 + iz^3 \quad \forall z \in \mathbb{R}$$

Fie $g: \mathbb{C} \rightarrow \mathbb{C}$, $g(z) = z^4 + \frac{i}{4}z^4 \quad \forall z \in \mathbb{C}$

g funcție omerică și $g'(z) = 4z^3 + iz^3 \quad \forall z \in \mathbb{C}$

Observăm că $f'(z) = g'(z) \quad \forall z \in \mathbb{R}$

f, g omerice $\Rightarrow f', g'$ funcții omerice } \begin{matrix} \text{teorema} \\ \Rightarrow \text{de} \\ \text{identitate} \end{matrix}

$$\Rightarrow f'(z) = g'(z) \quad \forall z \in \mathbb{C} \quad \Rightarrow (f-g)' = 0 \quad \Rightarrow$$

$$\Rightarrow \exists c \in \mathbb{C} \text{ a.c. } f-g \equiv c, \text{ adică } f(z) = g(z) + c \quad \forall c \in \mathbb{C}$$

$$\Rightarrow f(z) = z^4 + \frac{i}{4}z^4 + c, \quad \forall z \in \mathbb{C} \quad \Rightarrow c \in \mathbb{C}$$

$$\Rightarrow f(z) = (x+iy)^4 + \frac{i}{4}(x+iy)^4 + c$$

$$(x+iy)^4 = x^4 - y^4 + 2ix^3y$$

$$(x+iy)^4 = [(x+iy)^2]^2 = (x^2 - y^2 + 2ixy)^2 =$$

$$= x^4 + y^4 - 4x^2y^2 - 2x^2y^2 - 4ix^3y + 4ix^3y$$

$$\Rightarrow f(z) = x^4 + y^4 - 6x^2y^2 - 4ix^3y + 4ix^3y + \frac{i}{4}x^4 + \frac{i}{4}y^4 - ix^3y - \frac{i}{2}x^2y^2 + xy^3 - x^2y + c$$

$$= (x^4 + y^4 - 6x^2y^2 + xy^3 - x^2y + i(-4xy^3 + 4x^3y + \frac{x^4}{4} + \frac{y^4}{4} - x^3y - \frac{1}{2}x^2y^2)) + c$$

$$\operatorname{Re} f = x^4 + y^4 - 6x^2y^2 + xy^3 - x^2y = u \quad \left| \quad \begin{matrix} a) \operatorname{Re}(c) = 0 \\ \operatorname{Im}(c) = b \in \mathbb{R} \end{matrix} \right.$$

$$\text{Deci } f(z) = (x+iy)^4 + \frac{i}{4}(x+iy)^4 + bi \quad \forall z \in \mathbb{C}, b \in \mathbb{R}$$

Exercițiul 2:

$$I = \int_{-\infty}^{\infty} \frac{z^4 - 1}{z^4 + 2z^2 + 1} dz$$

$$z^4 + 2z^2 + 1 = 0$$

$$\text{Notăm } t = z^2 \Rightarrow t^2 + 2t + 1 = 0 \Rightarrow (t+1)^2 = 0 \Rightarrow$$

$$\Rightarrow t+1=0 \Rightarrow t=-1 \Rightarrow z^2=-1 \Rightarrow z \in \{-i, i\}$$

$$\text{Fie } f: \mathbb{C} \setminus \{\pm i\} \rightarrow \mathbb{C}, \quad f(z) = \frac{z^4 - 1}{z^4 + 2z^2 + 1}$$

Observăm că $f \in H(\mathbb{D})$.

$i, -i$ sunt puncte singular izolate pentru f (1)

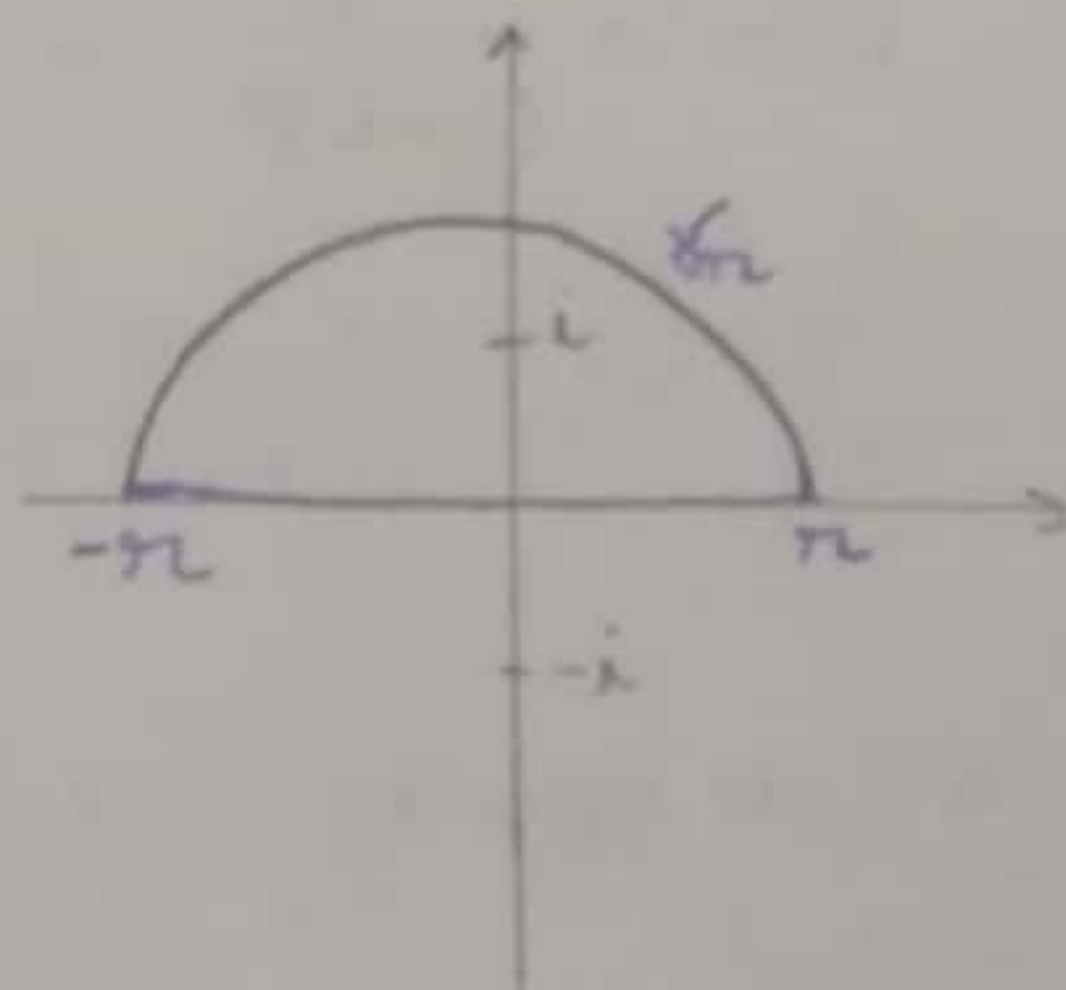
$$\mathbb{C} = (\mathbb{C} \setminus \{\pm i\}) \cup \{i, -i\}$$

Pentru $n > 1$ ($n > |i|$)

$$\Gamma_n = [-n, n] \cup \gamma_n,$$

unde $\gamma_n: [0, \pi] \rightarrow \mathbb{C}$,

$$\gamma_n(t) = n \cdot e^{it}$$



$\Rightarrow \Gamma_n$ închis, de clasă C^1

pe porțiuni (adică rectificabil),

$$\text{Im } \Gamma_n \subseteq \mathbb{C} \setminus \{\pm i\} \text{ și } \Gamma_n \sim 0 \quad (2)$$

$$\text{Din (1) și (2) teorema reziduurilor \quad \int_{\Gamma_n} f(z) dz = 2\pi i \left(\sum I(\Gamma_n, -i) \right)$$

$$\cdot \text{Res } f(-i) + I(\Gamma_n, i) \cdot \text{Res } f(i)$$

$$i \in \text{Int } \Gamma_n \Rightarrow J(\Gamma_n, i) = 1$$

$$i \notin \text{Int } \Gamma_n \Rightarrow J(\Gamma_n, -i) = 0$$

$$\text{Deci } \int_{\Gamma_n} f(z) dz = 2\pi i \cdot \text{Res } f(i)$$

Calculăm $\text{Res } f(i)$.

$$f(z) = \frac{z^4 - 1}{z^4 + 2z^2 + 1} = \frac{z^4 - 1}{(z-i)^2(z+i)^2} = \frac{\frac{z^4 - 1}{(z+i)^2}}{(z-i)^2} \quad \forall z \in \mathbb{D}$$

$$\left. \begin{aligned} \text{Considerăm } g: \mathbb{C} \setminus \{-i\} \rightarrow \mathbb{C}, \quad g(z) &= \frac{z^4 - 1}{(z+i)^2} \\ g &\in H(\mathbb{C} \setminus \{-i\}) \end{aligned} \right\} \Rightarrow$$

$$f(z) = \frac{g(z)}{(z-i)^2} \quad \forall z \in \mathbb{D}$$

$$g(i) = \frac{i^4 - 1}{(i+i)^2} = \frac{i^4 - 1}{(2i)^2} = \frac{-1 - 1}{-4} = \frac{2}{4} = \frac{1}{2} \neq 0$$

$\Rightarrow i$ pol de ordin 2

$$\text{Res } f(i) = \lim_{z \rightarrow i} \frac{1}{(2-1)!} [(z-i)^2 f(z)]^{(2-1)} =$$

$$= \lim_{z \rightarrow i} \left((z-i)^2, \frac{z^4 - 1}{z^4 + 2z^2 + 1} \right)' = \lim_{z \rightarrow i} \left(\frac{z^4 - 1}{(z+i)^2} \right)' =$$

$$= \lim_{z \rightarrow i} \frac{2z(z+i)^2 - (z^4 - 1) \cdot 2(z+i)}{(z+i)^4} =$$

$$= \frac{2i(2i)^2 - (-2) \cdot 2 \cdot 2i}{(2i)^4} = \frac{(2i)^3 + 8i}{16i^4} = \frac{8i^3 + 8i}{16i^4} = \frac{-8i + 8i}{16} = 0$$

$$\text{Totodată } \int_{\Gamma_n} f(z) dz = \int_{[-n, n]} f(z) dz + \int_{\gamma_n} f(z) dz$$

$$\Rightarrow 2\pi i \cdot \text{Res } f(i) = \int_{[-n, n]} f(z) dz + \int_{\gamma_n} f(z) dz$$

$$\int_{[-n, n]} f(z) dz = \int_{-n}^n f(x) dx, \text{ unde } \gamma: [-n, n] \rightarrow \mathbb{R}, \gamma(t) = t$$

$$\left| \int_{\gamma_n} f(z) dz \right| \leq l(\gamma_n) \cdot \sup_{z \in \gamma_n} |f(z)|$$

$$l(\gamma_n) = \frac{2\pi n}{2} = \pi n$$

$$|f(z)| = \left| \frac{z^4 - 1}{z^4 + 2z^2 + 1} \right| = \frac{|z^4 - 1|}{|z^4 + 2z^2 + 1|} \leq \frac{|z^4 - 1|}{|z^4| - 2|z^2| - 1} =$$

$$= \frac{|z^4 - 1|}{|z|^4 - 2|z|^2 - 1} \leq \frac{|z|^4 + 1}{|z|^4 - 2|z|^2 - 1} = \frac{n^4 + 1}{n^4 - 2n^2 - 1}$$

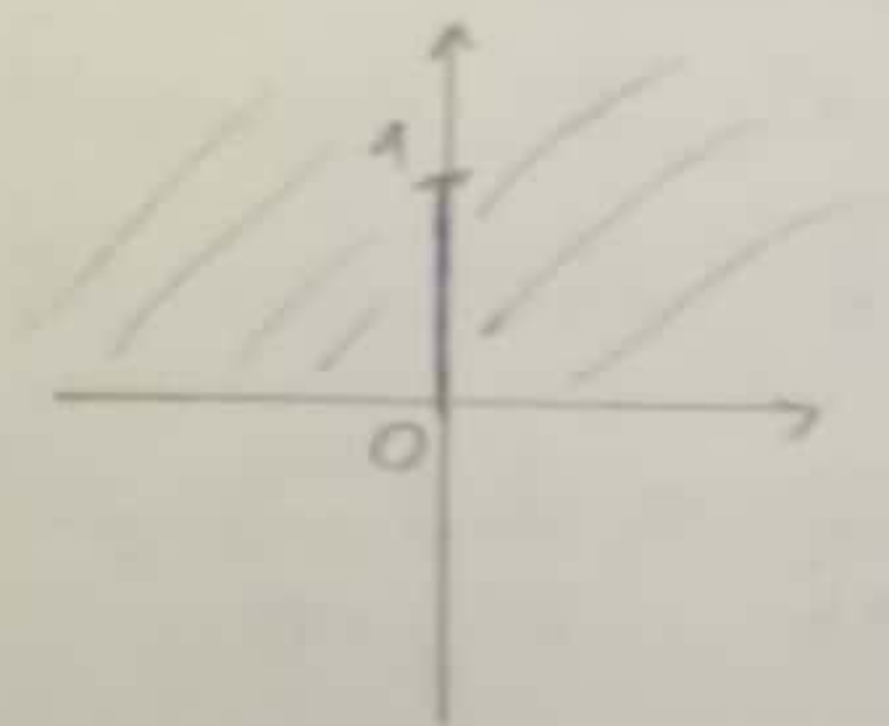
$$(\forall z \in \gamma_n \Rightarrow |z| = n)$$

$$\text{Deci } \left| \int_{\gamma_n} f(z) dz \right| \leq \pi n \cdot \frac{n^4 + 1}{n^4 - 2n^2 - 1} = \pi \cdot \frac{n^3 + n}{n^4 - 2n^2 - 1} \xrightarrow{n \rightarrow \infty} 0$$

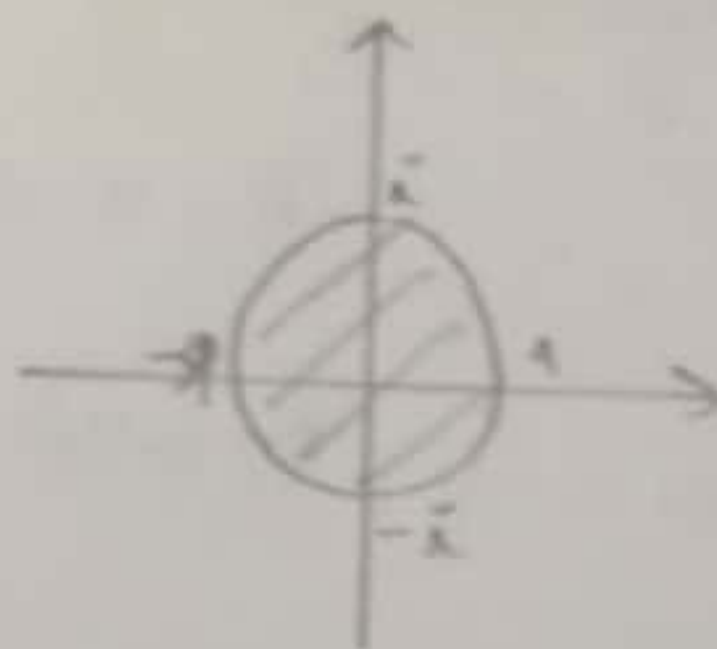
$$\underbrace{2\pi i \cdot \text{Res } f(i)}_{=0} = \underbrace{\int_{[-n, n]} f(z) dz}_{\downarrow n \rightarrow \infty} + \underbrace{\int_{\gamma_n} f(z) dz}_{\downarrow 0} \Rightarrow i = 0 \Rightarrow$$

$$\Rightarrow \int_{-\infty}^{\infty} \frac{x^4 - 1}{x^4 + 2x^2 + 1} dx = 0$$

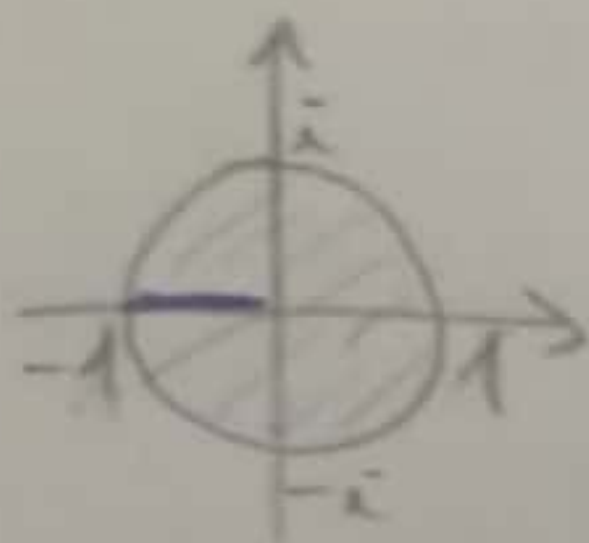
Exercitiul 3:



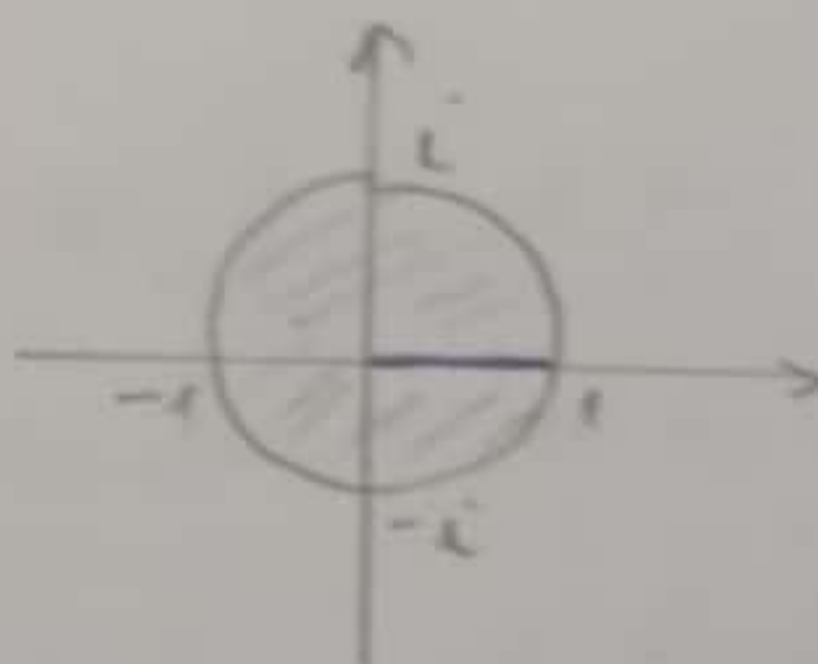
$$f = ?$$



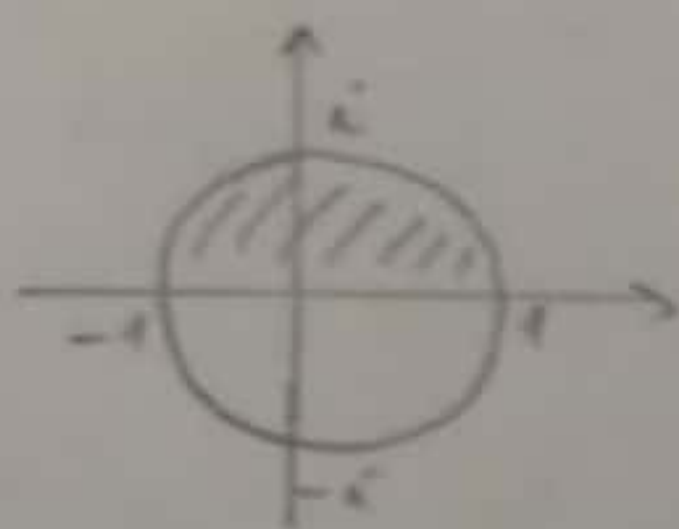
$$f_1(z) = \frac{z-i}{z+i}$$



$$f_2(z) = z \cdot e^{i\pi}$$



$$f_3(z) = z^{\frac{1}{2}}$$



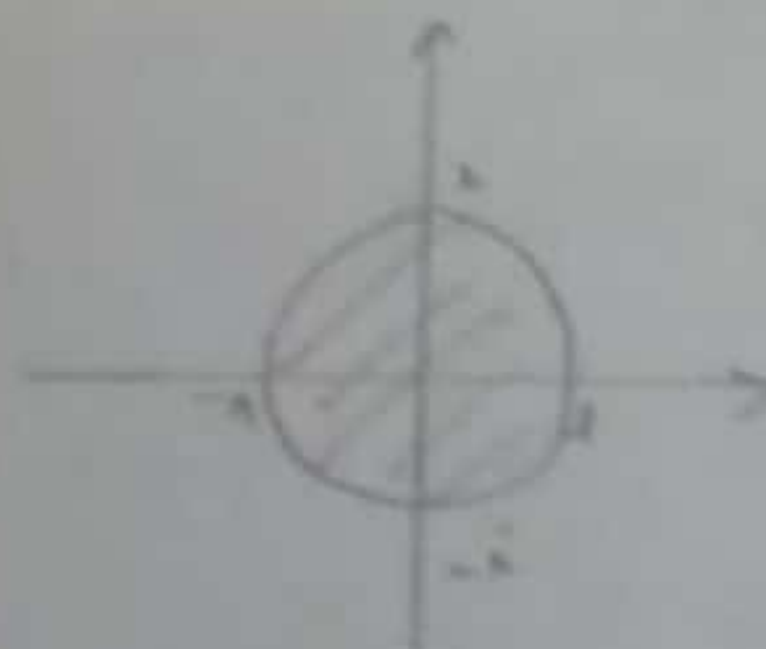
$$f_4(z) = i \cdot \frac{z+1}{1-z}$$



$$f_5(z) = -z^2$$



$$f_6(z) = \frac{z-i}{z+i}$$



$$\Omega = \mathbb{H}^+ \quad z \in \mathbb{C}$$

$$f_1: \Omega \rightarrow D_1, \quad f_1(z) = \frac{z-i}{z+i}$$

$$D_1 = \mathbb{D}(0, 1) \setminus [-1, 0]$$

$$\cancel{f_1(0)} = f_1(iy) = \frac{iy-i}{iy+i} = \frac{i(y-1)}{i(y+1)} =$$

$$= \frac{y-1}{y+1}, \quad \text{unde } y \in [0, 1]$$

$$f_1(0) = -1, \quad f_1(1) = 0$$

$$f_1\left(\frac{1}{2}\right) = \frac{-\frac{1}{2}}{\frac{3}{2}} = -\frac{2}{3}$$

$$f_2: D_1 \rightarrow D_2, \quad f_2(z) = z \cdot e^{i\pi}$$

$$f_3: D_2 \rightarrow D_3, \quad f_3(z) = z^{\frac{1}{2}}$$

$$f_4: D_3 \rightarrow D_4, \quad f_4(z) = i \frac{z+1}{1-z}$$

$$f_5: D_4 \rightarrow D_5, \quad f_5(z) = -z^2$$

$$f_6: D_5 \rightarrow D_6, \quad f_6(z) = \frac{z-i}{z+i}$$

$f = f_6 \circ f_5 \circ f_4 \circ f_3 \circ f_2 \circ f_1$ compunere de fct, olomorfe, bijective

$$D_2 = D_1, \quad D_3 = \{z \in \mathbb{D}(0, 1) \mid \operatorname{Im} z > 0\}$$

$$D_4 = \{z \mid \operatorname{Im} z > 0, \operatorname{Re} z < 0\}$$

$$D_5 = \{z \mid \operatorname{Im} z > 0\}$$

$$D_6 = \mathbb{D}(0, 1)$$

Exercițiul 4:

Studiem nr. rădăcinilor ale funcției f în bila unitate.

Fie $g, h \in H(\mathbb{D})$, $g(z) = 3z^2$ și $h(z) = z^3 + bz + b^2$

$$|h(z)| = |z^3 + bz + b^2| \leq |z|^3 + |b| |z| + |b|^2 =$$

$$= |z|^3 + |b| \cdot |z| + |b|^2 = 1 + |b| + |b|^2 < 1 + 1 + 1^2 = 3$$

$\uparrow$
 $|b| < 1$
 (ipoteză)

$|g(z)|$
 $\forall z \in B(0, 1)$

$$|g(z)| = |3z^2| = 3 \cdot |z^2| = 3$$

Deci $\gamma = B(0, 1)$ drum închis, rectificabil

$$\gamma \sim 0$$

$$|h(z)| < |g(z)|$$

$\Rightarrow$

~~teorema~~
 $\Rightarrow$ ~~Rouche~~ g și $\overbrace{g+h}^{=f}$ au același număr de rădăcini

în bila unitate

$$g(z) = 0 \Rightarrow 3z^2 = 0 \Rightarrow 2 \text{ rădăcini} \Rightarrow$$

$$\Rightarrow f \text{ are două rădăcini în } B(0, 1) \Rightarrow$$

$$\Rightarrow f \text{ are } 3 - 2 = 1 \text{ rădăcini în set } B(0, 1).$$